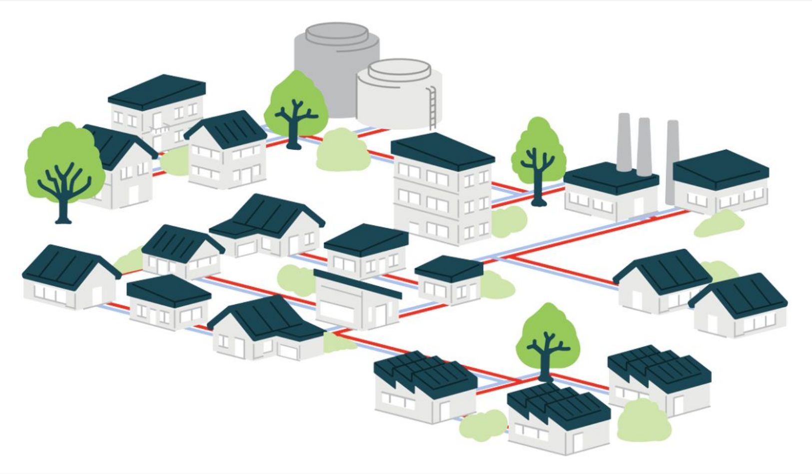
Electricity Consumption Prediction

Machine-Learning-Project



ZHAW – CAS Machine Intelligence – Machine Learning

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Data Used

Category	Description
Source	HEAPO Dataset(1,408 households in Zurich, recorded at 15-minute and daily resolutions)
Records	Monthly household observations
Features	Time, weather, household characteristics
Target	Electricity consumption (kWh)

01

Smart meter → monthly consumption

02

Weather → average temperaturen

03

Metadata → household characteristics

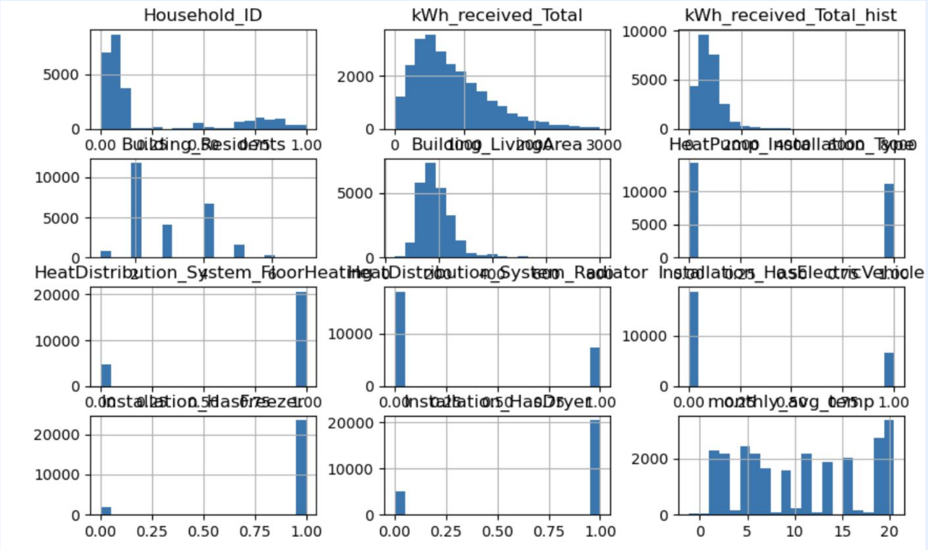
04

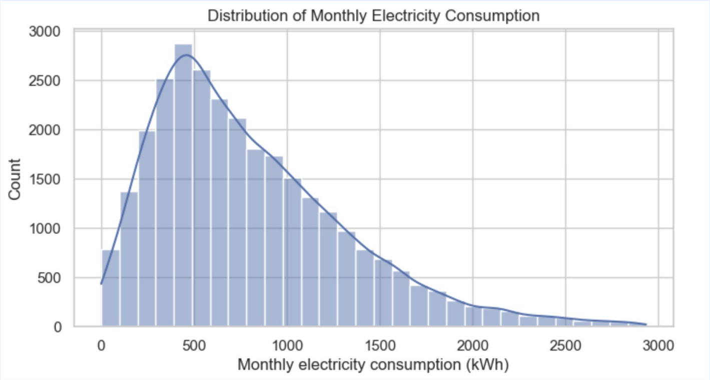
Merged into unified dataset

Data Process

Exploratory Data Analysis:

- Electricity consumption is right-skewed
  → most households use moderate energy, few use very high
- Larger homes tend to consume more
  → but relationship is weak and noisy
- Lower temperature → higher consumption
  → reflects heating demand in colder periods





Data Process

Data Preparation Process:

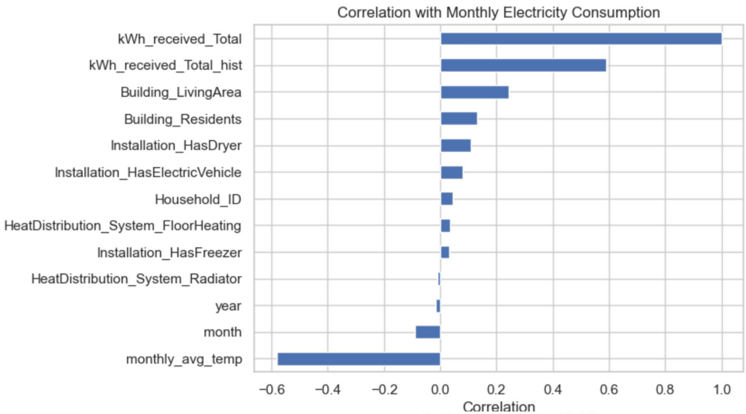
- Aggregated to monthly level
- Merged datasets on time + household
- Handled missing values
- Removed extreme outliers

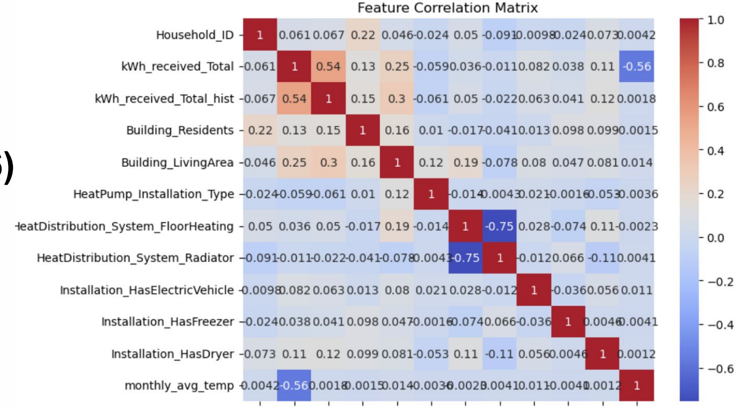
Before	After
22.8M raw records	29,254 monthly records
High-frequency time series	Aggregated per household & month
Missing values present	0 missing values (critical columns)
Separate datasets	1 unified dataset

Feature Selection

Selected relevant predictors:

- Temperature → heating demand
- Household → size & usage behavior
- Time → seasonal patterns





Strong negative correlation with temperature **(~ -0.56)**
→ lower temperature leads to higher electricity consumption
Historical consumption strongly correlated **(~ 0.54)**
→ past usage is a key predictor

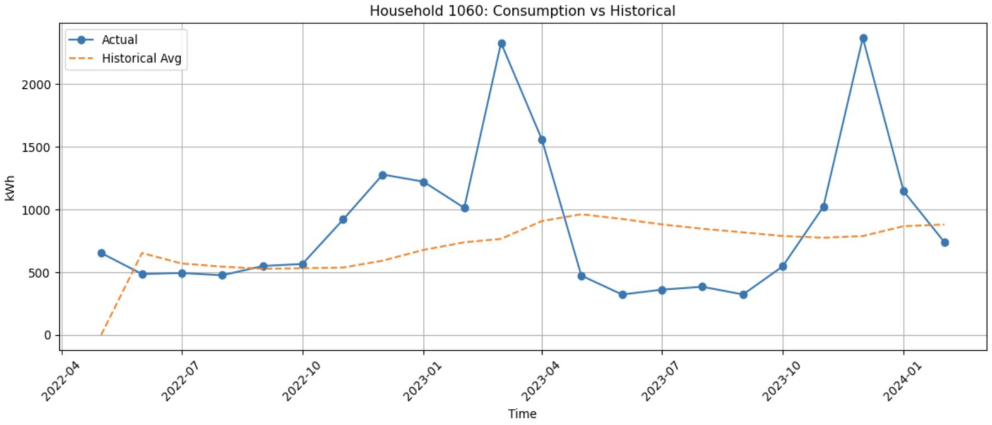
Feature Engineering Approach

Time-Based Features

- **Month** → captures yearly consumption patterns
- **Season** → groups months into heating

  vs non-heating periods

👉 Energy usage varies strongly across seasons



Historical Feature

- **Past Consumption (lag / rolling mean)**
  - Represents previous usage behavior
  - Captures household-specific patterns

👉 One of the most important predictors

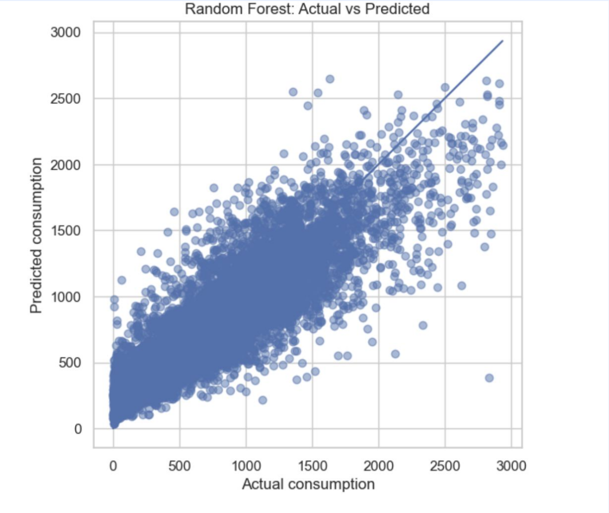
Modeling Approach

Regression task

- Linear Regression
- Random Forest
- Compare linear vs non-linear

Model	MAE	RMSE	R^2
Random Forest	183.06	252.32	0.75
Linear Regression	198.70	270.20	0.71

Random Forest: Actual vs Predicted
Shows good fit but some variance

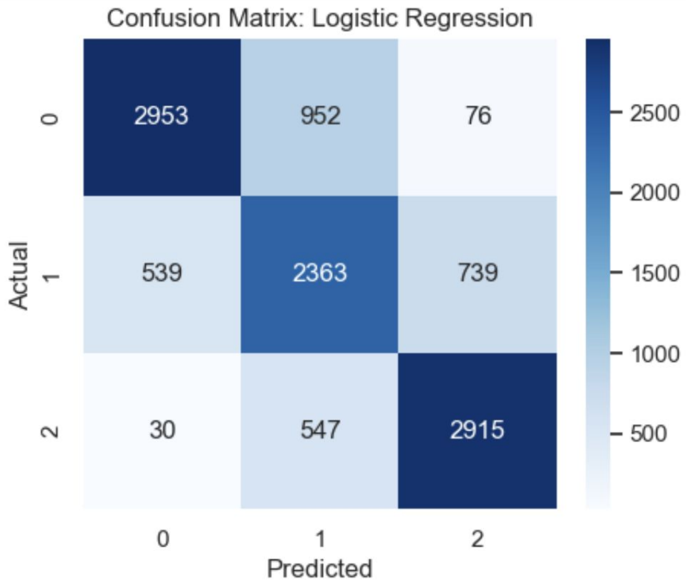


Classification Model Performance

We convert consumption into:

low (0) **medium** (1) **high** (2)

- Accuracy ~52%
- F1 score ~0.49
- Slightly better than random
- Not suitable for this problem



Insights and Conclusions

- Strong seasonal patterns
- Temperature impacts consumption
- Random Forest performs best
- $R^2 \approx 0.75 \longrightarrow$ strong predictive power
- Problem still complex